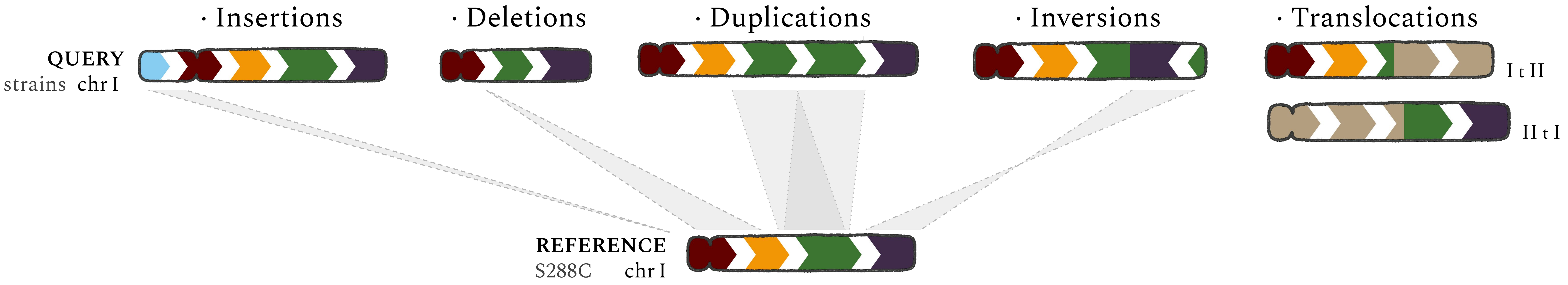
# Impact of Structural Variants on Genome and Proteome Evolution in *Saccharomyces cerevisiae*


Andrea TARABINI [1,2], Gilles FISCHER [1], Anne LOPES [2]

1. Sorbonne Université, CNRS, UMR7238, CQSB - Computational, Quantitative and Synthetic Biology, 75005 Paris, France
2. Université Paris-Saclay, CNRS, UMR9198, I2BC - Institute for Integrative Biology of the Cell, 91190 Gif-sur-Yvette, France


## INTRODUCTION

**Structural Variants** (**SVs**) are genomic rearrangements affecting DNA on a large scale. SVs can be broadly classified into:




SVs can **alter gene dosage**, **disrupt coding sequences**, or **create novel gene fusions**; hence they likely have **functional** and **evolutionary impacts**. The advent of Long-Read Sequencing (LRS) Technologies has alleviated the historical limitations in SVs detection.
The ***Saccharomyces cerevisiae*** **Reference Assembly Panel** (**ScRAP** [1]) enables unraveling the Structural Landscape at the **population** level.

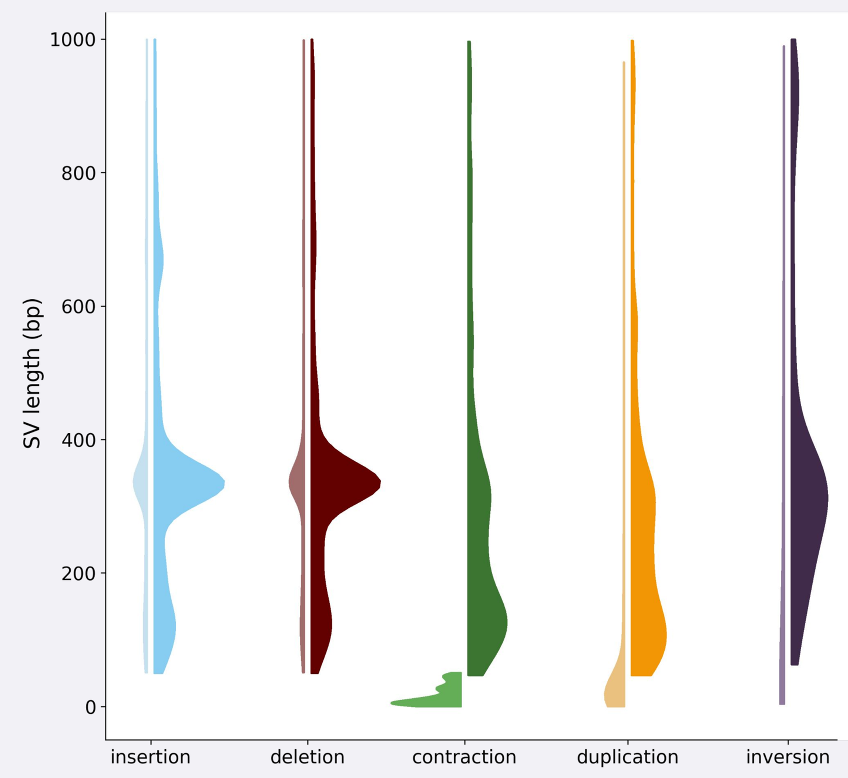
## MUM&Co

An **assembly-based** SV calling method capable of accurately detecting virtually all classes of genomic variants [2]. It is based on the **MUMmer nucmer** algorithm and **dot-plot alignment** principles [3].

We have recently expanded its SVs detection capabilities via a revision of the original bash script and a python extension script. The method has been enhanced to enable the detection of interspersed duplications and contractions and has been revised overall to improve **robustness** and **interpretability**.

When applied to **183** ***S. cerevisiae*** **genome assemblies**, the latest version of MUM&Co identified **60'028 SVs**, with a median of 336 SVs per assembly.

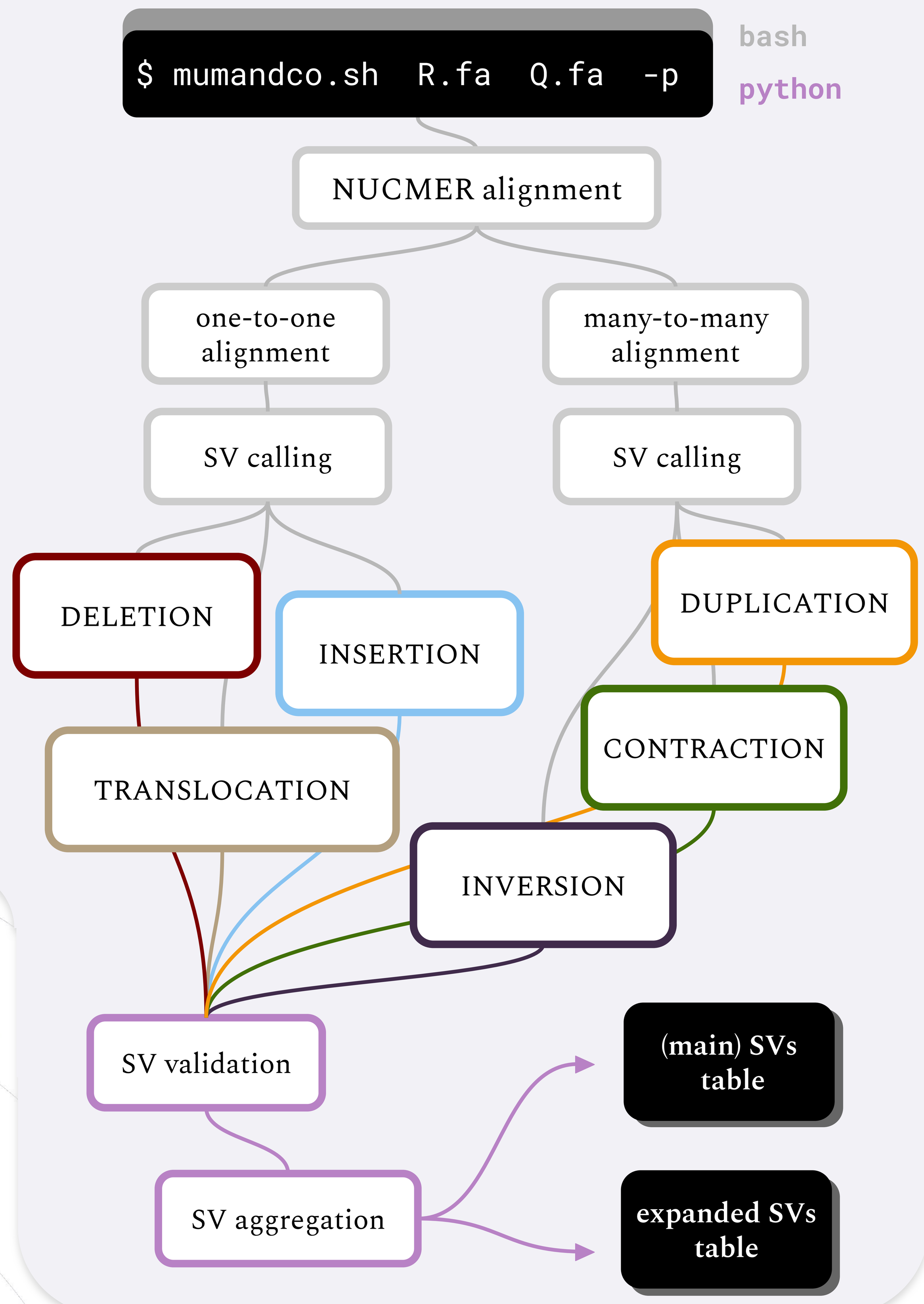
**Detail view of SVs length distribution: MCO39 vs MCO4**




**MEDIAN SV LENGTH** (bp)

342 | 340 433 | 383 10 | 301 28 | 319 13307 | 334

**MUM&Co overview**



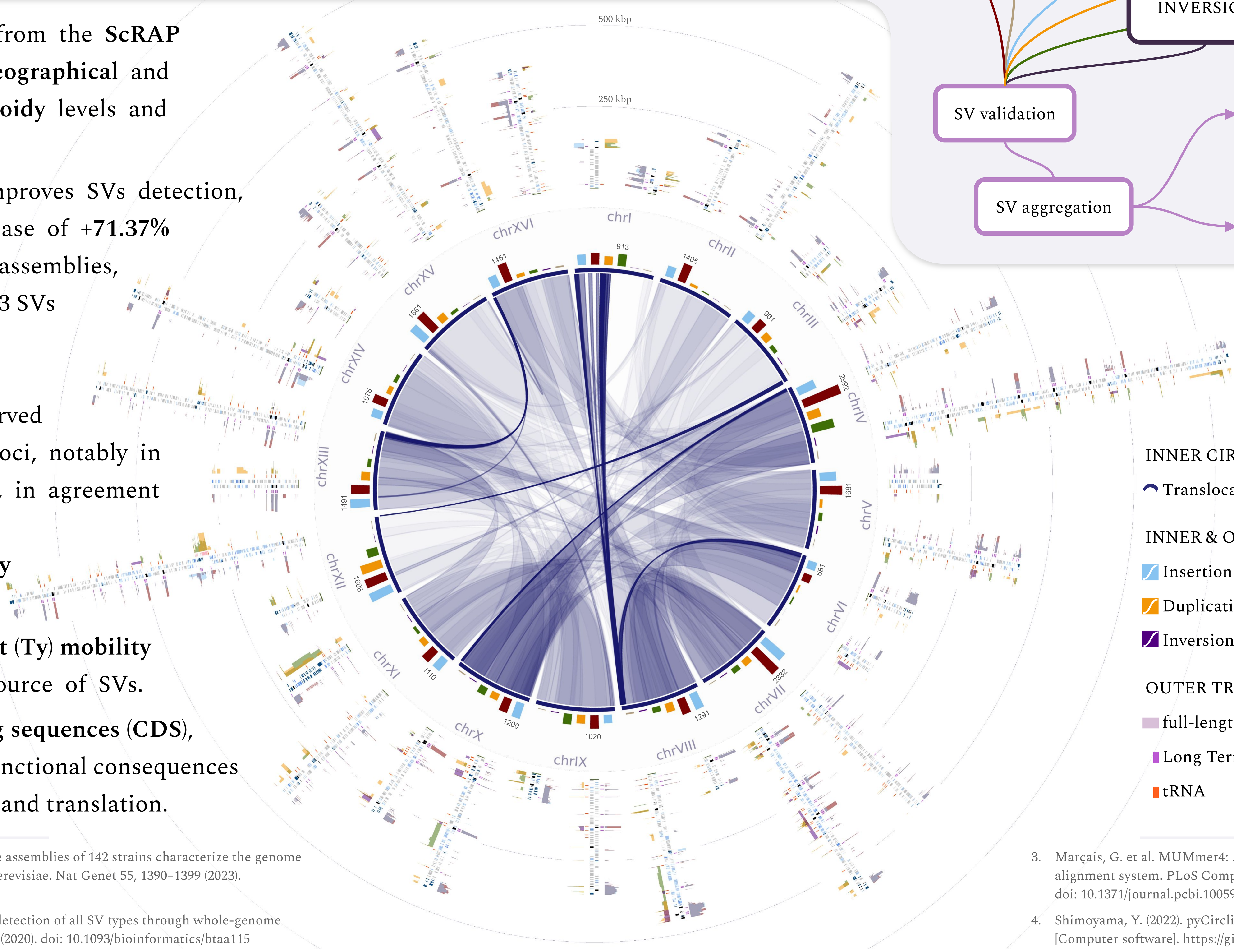

The strains sourced from the **ScRAP** encompass diverse **geographical** and **ecological origins**, **ploidy** levels and **heterozygosity**.

**Assembly phasing** improves SVs detection, with an average increase of **+71.37%** relative to collapsed assemblies, corresponding to +253 SVs per assembly.

A tendency for SVs accumulation is observed at specific genomic loci, notably in **subtelomeric regions**, in agreement with the elevated **evolutionary plasticity** of these regions.

**Transposable element** (**Ty**) **mobility** constitutes a major source of SVs.

SVs affect **gene coding sequences** (**CDS**), indicating potential functional consequences for gene transcription and translation.




INNER CIRCLE:
Translocation

INNER & OUTER TRACK:
Insertion Deletion
Duplication Contraction
Inversion Translocation

OUTER TRACK:
full-length Ty CDS <2 ≤5 ≤10 ≤50 >50
Long Terminal Repeats
tRNA rRNA centromere


1. O'Donnell, S. et al. Telomere-to-telomere assemblies of 142 strains characterize the genome structural landscape in Saccharomyces cerevisiae. Nat Genet 55, 1390–1399 (2023). doi: 10.1038/s41588-023-01459-y
2. O'Donnell, S. et al. MUM&Co: accurate detection of all SV types through whole-genome alignment. Bioinformatics 36, 3242–3243 (2020). doi: 10.1093/bioinformatics/btaa115
3. Marçais, G. et al. MUMmer4: A fast and versatile genome alignment system. PLoS Comput Biol 14, e1005944 (2018). doi: 10.1371/journal.pcbi.1005944
4. Shimoyama, Y. (2022). pyCirclize: Circular visualization in Python [Computer software]. https://github.com/moshi4/pyCirclize